AUDIOLIT — INTERPRETABILITY WORKBENCH

Dataset Ingestion & Prediction Pipeline — Demo Guide

A verified, click-by-click demo script covering all 7 benchmark corpora and the custom-dataset path, with an honest working/not-working breakdown per dataset.

SCOPE
Backend dataset ingestion (LIT-123/141/142/181/236/235) + custom dataset ground truth (LIT-247) + prediction display fix (LIT-249)

VERIFIED AGAINST
Live repo state on this machine (code + `Backend/data/` contents), not documentation claims

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1. Executive summary — what actually works right now

This isn't a description of what the code is *supposed* to do — every row below was checked against the real files under `Backend/data/` and the live routing logic, not just the ticket acceptance criteria. Two things are broken or incomplete in ways worth knowing before you demo them live.

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FULLY WORKING CORPORA	PARTIAL (MISSING GROUND TRUTH)	NOT DEMOABLE LOCALLY	KNOWN CACHING BUG (BATCH)

SAFE TO DEMO END-TO-END, NO CAVEATS

Common Voice (ASR), ASVspoof 2021 DF (Deepfake), CREMA-D (SER), RAVDESS (SER), ESD (SER) — all have real audio + labels on disk, a working loader, and show up correctly in the UI.

DEMOABLE BUT SHOW A KNOWN GAP ON PURPOSE

L2-ARCTIC — audio plays and the accent/L1 metadata is correct, but the **Ground Truth column will be blank for every row**: the transcript files the loader expects aren't present in this machine's local data folder (only the `wav/` subfolder was provisioned, not `transcript/`). Good to demo the accent-bias angle; don't promise a ground-truth transcript will appear.

DO NOT ATTEMPT TO SELECT THIS LIVE

LibriSpeech — the loader code is complete and unit-tested (LIT-141), and it's correctly registered, but **no LibriSpeech data exists anywhere in this repo's `Backend/data/`** (confirmed: no such folder on disk). The frontend already knows this — it calls the live `/datasets/list` availability check and greys the option out with **"(not provisioned)"** in the dropdown, so a demo audience will just see it disabled rather than the app crashing. If asked, explain it as "built and tested, just not loaded onto this laptop."

KNOWN, PRE-EXISTING, UNFIXED — BATCH CACHING ONLY

Batch inference over SER/Deepfake datasets (RAVDESS, CREMA-D, ESD, ASVspoof) will look slower on repeat runs than it should, because the pre-flight cache-check endpoint (`POST /inferences/batch-check`) uses a different cache-key format than the endpoint that actually stores results, so it essentially never finds a hit for those model types. **Actual predictions are still correct** — this only affects how fast a repeated batch run feels, not accuracy. See §5.

2. Per-dataset status — verified against the real files on disk

Dataset	Task	Loader	Data on disk	Ground truth	Frontend	Demo verdict
Common Voice	ASR	Working	100 audio files	Yes (sentence)	Selectable	Best default ASR demo — the original reference loader.
LibriSpeech	ASR	Working (code)	None found	N/A locally	Disabled — "not provisioned"	Do not select live. Explain as tested-but-unprovisioned.
ASVspoof 2021 DF	Deepfake	Working	250 audio files + protocol file	Yes (bona-fide / spoof)	Selectable	Best deepfake demo — also proves the LIT-249 label-display fix.
L2-ARCTIC	ASR	Working	24 speakers, audio only	Missing (no transcript/ folder)	Selectable	Fine for accent-bias story; Ground Truth column will read blank.

Dataset	Task	Loader	Data on disk	Ground truth	Frontend	Demo verdict
CREMA-D	SER	Working	120 audio files, flat layout	Yes (from filename)	Selectable	Works via the loader's flat-directory fallback (no `AudioWAV/` subfolder present). Demographic breakdown (age/sex/race) is empty — `VideoDemographics.csv` isn't provisioned.
RAVDESS	SER	Working	144 audio files, flat layout	Yes (from filename)	Selectable	Works via the loader's flat-directory fallback (no `Actor_*/` subfolders present).
ESD	SER	Working	100 audio files + catalog CSV	Yes (normalized emotion labels)	Selectable	Good demo for the LIT-236 BOM/label-normalization fix story.
Custom dataset (any name)	Any	Working	User-uploaded	Optional, via CSV upload	Selectable after creation	Best feature to show LIT-247's order-independent ground-truth matching.

"Data on disk" was checked directly (`ls` / `find` against `Backend/data/` on this machine) — a teammate's machine could differ if they've provisioned LibriSpeech or the missing L2-ARCTIC transcripts locally. Re-check before relying on this table on a different machine.

3. Full demo workflow — step by step

Part A — Built-in benchmark corpora (5–7 minutes)

1. **Open the Toolbar dataset dropdown**
   Point out that every entry came from a live `GET /datasets/list` call, not a hardcoded list — this is the LIT-235 fix. Any dataset without local data shows **"(not provisioned)"** and is greyed out (LibriSpeech will look like this).

2. **Select model: Whisper Base (ASR) → dataset: Common Voice**
   Run inference on 2–3 rows. Show the transcript appearing in the table and confirm it matches the Ground Truth column.

3. **Switch model to a Deepfake model (MelodyMachine or Wav2Vec2 XLSR) → dataset auto-switches to ASVspoof 2021**
   Run a batch of rows. **This is the LIT-249 fix** — point out the Predicted Label column shows a clean `bona-fide` / `spoof` value, not a JSON blob. Optionally open the sidebar's "Deepfake Detection Results" card to show the full probability breakdown lives there instead.

4. **Switch model to Wav2Vec2 (SER) → dataset: ESD or RAVDESS**
   Run inference, show emotion labels matching between Ground Truth and Predicted Label — this proves the LIT-236 label-normalization work (raw "Surprise" → "surprised") actually lines up with the model's own vocabulary.

5. **Optional: select L2-ARCTIC (still ASR-compatible)**
   Use this to demo accent metadata (speaker's native language shown per row) — but call out in advance that the Ground Truth column will be blank here, so it doesn't look like a live bug.

6. **Optional: try to select LibriSpeech**
   Show it's visibly disabled with "(not provisioned)" rather than being missing entirely or crashing — a good example of the app failing gracefully instead of silently.

Part B — Custom dataset + ground-truth CSV upload (3–4 minutes)

1. **Open Custom Dataset Manager → Create Dataset**
   Name it something demo-friendly, e.g. `demo-set`.

2. **Upload 3–5 audio files under the "Upload Files" tab**
   Show the dataset table — Ground Truth column is blank for every row (expected, by design).

3. **Switch to the "Ground Truth CSV" tab, upload a CSV**
   Columns can be named flexibly (`filename` / `file` / `audio_file` + `ground_truth` / `transcript` / `label` / `text`). Show the match-summary response: matched count / total, and any unmatched rows called out by name.

4. **Go back to the dataset table**
   Ground Truth column is now populated — with zero changes to the table component itself, proving the fix was entirely in how the data got there, not in the table's rendering.

5. **Optional: re-upload a different CSV**
   Show that it replaces rather than merges with the previous mapping — a deliberate design decision from the ticket.

4. Anticipated Q&A

Question	Answer
Why can't I select LibriSpeech?	The code to load it is written, tested, and registered — it just isn't provisioned on this machine (no data downloaded). The dropdown deliberately disables anything unprovisioned instead of letting you pick it and hit a 404.
Why is L2-ARCTIC's Ground Truth column empty?	The loader looks for a per-utterance transcript file next to each audio clip; those transcript files weren't included when this machine's L2-ARCTIC subset was provisioned (only the audio was). The accent/speaker metadata is still real and correct.
If I run the same deepfake batch twice, why doesn't it look faster the second time?	A separate pre-check endpoint that's supposed to short-circuit already-cached results uses a slightly different cache key format than the one that actually stores results, so for non-ASR models that pre-check basically never finds anything. The underlying prediction caching (used by per-row Regenerate) still works correctly — this is a batch-run performance quirk, not a correctness bug.
What was actually broken in the deepfake JSON display bug?	Only the summary table cell's formatting. The model's output was always correct, and the sidebar detail card always displayed it correctly — only the compact table cell fell back to dumping raw JSON because the code handling it only knew how to unwrap Whisper's plain-text response shape.
Can a user upload ground truth before uploading any audio?	Yes — by design, matching is order-independent. The CSV can be uploaded first, last, or in the middle of adding audio files, and the same end result occurs either way.
Does every SER/ASR corpus provide demographic bias data (age, gender, race)?	Not uniformly. Common Voice and CREMA-D are designed to carry it, but on this machine CREMA-D's demographics file isn't provisioned, so that field will show empty even though the loader supports it.

5. Known limitation — batch cache-check key mismatch (not fixed, flagged not blocking)

Documented here for completeness since it can look like a bug live if someone runs a large batch twice and expects a speed-up:

- `POST /inferences/batch-check` builds its lookup key as `f"{model}_{file_content_hash}"`
- The endpoint that actually performs and caches inference (`run_inference`) stores under `f"v2_{model}_{file_path_hash}"` — a different prefix
- Result: the pre-flight "is this already cached?" check almost never finds a hit, so batch runs recompute live even on a repeat pass
- Separately, the pre-check's transcript extractor (`as_transcript`) only understands ASR-shaped results — for an ADD/SER prediction shaped like `{"predicted_label": ...}` , it would return an empty string even on a rare hit

This was identified and deliberately left unfixed during the LIT-249 display-bug fix, as out of scope for that ticket (different endpoint, different bug class). Per-row Regenerate and the actual prediction values are unaffected — only the perceived speed of a repeat batch run.